Master Kubernetes Learning Tracker (Version 1.2 FINAL)

Student: Karthik
Course: Kubernetes from Beginner to Production (Custom ChatGPT Curriculum)
Primary Goal: Become a Production-Ready Kubernetes & AKS Engineer

Current Progress

■ Completed up to **Day 12A – Services Part 1**
➡ Next Session: **Day 12B – Services Part 2**

Phase 1 – Kubernetes Fundamentals

Day	Topic	Status
1	Kubernetes Architecture	■
2	Pods & YAML	■
3	ReplicaSets	■
4	Deployments	■
5	Scaling & Rollback	■
6	Labels & Selectors	■
7	Deployment Deep Dive	■

Phase 2 – Workload Management

Day	Topic	Status
8	Rolling Updates	■
9	Deployment Deep Dive	■
10	Deployment Scenarios & Interview Questions	■
11	Rolling Update Strategy	■
12A	Services Part 1	■
12B	Services Part 2	■
13A	Ingress Part 1	■
13B	Ingress Part 2	■
14	ConfigMaps & Secrets	■

Phase 3 – Storage & Scheduling

Day	Topic	Status
15	Volumes	■
16	PV, PVC & StorageClass	■
17	Namespaces, Resource Quotas & Limits	■
18	Scheduler, NodeSelector, Affinity, Taints & Tolerations	■

Phase 4 – Advanced Workloads

Day	Topic	Status
19	Jobs & CronJobs	■
20	Health Probes	■
21	StatefulSets & Headless Services	■
22	DaemonSets	■

Phase 5 – Security & Packaging

Day	Topic	Status
23	RBAC	■
24	Service Accounts	■
25	Secrets Deep Dive	■
26	Helm	■
27	Network Policies	■

Phase 6 – Production Kubernetes

Day	Topic	Status
28	Logging	■
29	Monitoring	■
30	Autoscaling (HPA/VPA/Cluster Autoscaler/KEDA)	■
31	Troubleshooting Pods	■
32	Troubleshooting Nodes	■
33	Troubleshooting Networking	■

Phase 7 – Azure Kubernetes Service (AKS)

Day	Topic	Status
34	AKS Architecture	■
35	AKS Networking	■
36	AKS Storage	■
37	AKS Identity	■
38	AKS Security	■
39	AKS Upgrades & Node Pools	■
40	AKS Production Troubleshooting	■

Phase 8 – CI/CD Integration

Day	Topic	Status
41	Docker + Kubernetes Workflow	■

42	Azure DevOps Pipelines with Kubernetes	■
43	Terraform + AKS	■
44	GitOps (ArgoCD Concepts)	■

Phase 9 – Projects & Interviews

Day	Topic	Status
45	Production Deployment Project	■
46	Multi-Tier Application	■
47	Troubleshooting Lab	■
48	Mock Interview - Beginner	■
49	Mock Interview - Intermediate	■
50	Mock Interview - Advanced	■

Phase 10 – Kubernetes Expert Track

Day	Topic	Status
51	Advanced Networking Part 1	■
52	Advanced Networking Part 2	■
53	Advanced Networking Part 3	■
54	Service Mesh Part 1	■
55	Service Mesh Part 2	■
56	Service Mesh Part 3	■
57	Service Mesh Part 4	■
58	Gateway API, Operators, CRDs, Kustomize	■
59	Enterprise Security (cert-manager, External Secrets)	■
60	Enterprise AKS Capstone & Final Review	■

Teaching Method

For every topic always follow:
1. Explain the problem.
2. Explain why Kubernetes introduced the feature.
3. Explain internal working.
4. ASCII architecture diagrams.
5. Explain every YAML field.
6. Demonstrate kubectl commands.
7. Show expected outputs.
8. Production scenarios.
9. Troubleshooting.
10. Hands-on.
11. Interview Questions.
12. Quiz before moving to next topic.